Solution:

x	f	fx
0	211	0
1	90	90
2	19	38
3	5	15
4	0	0
Total	325	143

(i) Mean $\bar{x} = \frac{\Sigma fx}{\Sigma f} = \frac{143}{325} = 0.44$

(ii) $\lambda = \bar{x} = \quad = 0.44$

(iii) $P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!} = \frac{e^{-0.44}(0.44)^x}{x!}$

(iv) $P(0) \quad = \frac{e^{0.44} \times 0.44^0}{0!} = e^{-0.44} \quad = 0.6440$ (from the Poisson table)

(v) $F(0) \; = \; N \times P(0) \; = 325 \times 0.6440 = 209.43$

(vi) $F(x+1) = \frac{\lambda}{x+1} F(x)$

$F(1) \; = \; F(0+1) \; = \frac{0.44}{0+1} \times 209.43 = 92.15$

$F(2) \; = \; F(1+1) \; = \frac{0.44}{1+1} \times 92.15 = 20.27$

$F(3) \; = \; F(2+1) \; = \frac{0.44}{2+1} \times 20.27 = 2.973$

$F(4) \; = \; F(3+1) \; = \frac{0.44}{3+1} \times 2.97 = 0.33$

Result:

(1). Fitted Poisson distribution is $P(X = x) = \frac{e^{-0.44}0.44^x}{x!}$, $x = 0, 1, 2\ldots.$

(2). Expected frequencies are given below :

x	0	1	2	3	4	Total
Observed frequencies (O_i)	211	90	19	5	0	325
Expected Frequencies (E_i)	210	92	20	3	0	325

10.3.3 Fitting of Normal Distribution

In fitting a Normal distribution to the observed data, given in class intervals, we follow the following procedure:-

(i) Calculate μ and σ of the distribution